JAYAPRAKASH YADAV GUNTUMANI

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Summary

AI Engineer with 3 years of experience designing, deploying, and operating production machine learning systems. Skilled in real-time inference, backend engineering, and ML infrastructure, with hands-on experience building agentic LLM applications, RAG pipelines, and fine-tuned language models.

Projects

VersionPilot - AI-Powered Dependency Migration Platform Link

- Built an AI agent that analyzes Python repositories for dependency upgrade risks and deprecated API usage, generating source-linked migration plans with exact files, line numbers, and required code changes beyond Dependabot-style version alerts.
- Designed a 6-stage LangGraph workflow using tool calling, structured outputs, AST-based API scanning, LLM release-note extraction, critic validation, and recovery routing, achieving 100% precision, 96.7% recall, and 0.98 F1 for deprecated API detection.
- Built production safeguards that prevented misleading migration recommendations across 10/10 simulated API, repository, and LLM failure scenarios using confidence scoring, provenance tracking, and deterministic fallbacks.

Role Based Access Control (RBAC) Chatbot Link

- Built an RBAC-enforced RAG system over 5 departments, using metadata-filtered vector retrieval to guarantee role-scoped access and prevent cross-role document leakage, with LangChain, Gemini, and ChromaDB.
- Improved context recall 22% and answer relevancy 2.5× over a baseline retriever by combining hierarchical chunk merging with hybrid retrieval (vector + BM25), validated through RAGAS evaluation and tracked across experiments in MLflow.
- Deployed a Dockerized RAG service to GCP Cloud Run, serving cited responses with <3s p95 latency and 0.87+ RAGAS faithfulness.

Code-to-Comment Generator Link

- Fine-tuned CodeT5 with LoRA for automated docstring generation on 21K CodeSearchNet samples, reaching 0.73 BERTScore F1 (+15% over base) while cutting trainable parameters 95% - used AST parsing to strip in-code comments and prevent training-target leakage.

Professional Experience

Tesla Solutions — **Sep 2025 - May 2026**
Machine Learning Engineer (Intern) — *Houston, TX*

- Engineered a real-time production ML inference pipeline in C#/.NET with temporal feature engineering and ONNX Runtime model serving, enabling sub-second well-state predictions through native SharpWatch deployment.
- Raised multi-well operational state prediction accuracy from 85% to 95% by designing, training, and deploying a PyTorch LSTM through ONNX Runtime into production.
- Increased RAMPING-state recall from 71% to 96% on an imbalanced dataset by engineering temporal features such as lag, rolling statistics, and choke rate of change to capture early state transitions.
- Implemented production observability using structured .NET logging, Grafana dashboards for prediction metrics and inference failures, and Azure DevOps CI pipelines, which shortened issue detection time and streamlined deployment of ML updates.

Optum Global Solutions (UHG) — **May 2022 - Jul 2024**
Associate Data Scientist — *India*

- Developed a FastAPI backend with scheduled, Dockerized batch services to migrate thousands of Power BI subscription and bursting reports at scale.
- Designed a database-backed tracking layer that recorded per-batch migration status and surfaced recurring report failures, enabling targeted debugging and faster issue resolution.
- Reduced infrastructure costs by $200K annually by migrating data processing to the PySpark Pandas API, processing 5TB+ datasets 40% faster.
- Detected distribution shifts across 20+ production ML models before performance degradation by building automated data drift monitoring pipelines and Grafana dashboards.

Technical Skills

- **AI & LLM**: LangGraph, LangChain, RAG, Tool Calling, Structured Outputs, Prompt Engineering, LoRA
- **ML & Data**: PyTorch, Time-Series, NLP, PySpark, Pandas, ONNX Runtime
- **Backend & Cloud**: Python, FastAPI, C#/.NET, REST APIs, SQL, Docker, Kubernetes, Vertex AI, Cloud Run, Azure DevOps
- **Evaluation & Vector DBs**: LangSmith, RAGAS, MLflow, ChromaDB, FAISS

Education

University of Houston — **Aug 2024 - May 2026**
Master of Science, Engineering Data Science — *Houston, Texas*

- **GPA:** 3.967/4.0